

R for bioinformatics, data wrangler, part 2

HUST Bioinformatics course series

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Section 1

section 1: TOC

前情提要

pipe

- pipe

dplyr

- select()
- filter()
- mutate()
- summarise()
- arrange()
- group_by() ...

今次提要

tidyr

- ① 长宽数据转变
- ② 数据分割: 一列变多列
- ③ 数据合并: 多列变一列
- ④ 其它函数

Section 2

section 2: data wrangler - tidy

tidyr

what is tidyr ?

The goal of tidyr is to help you create **tidy** data.



Figure 1: dplyr logo

more to read

- [tidyr official page at tidyverse](#)
- [R for data science](#)

tidyr 安装

只需安装一次即可！

```
# The easiest way to get tidyr is to install the whole tidyverse:
```

```
install.packages("tidyverse")
```

```
# Alternatively, install just tidyr:
```

```
install.packages("tidyr")
```

```
# Or the development version from GitHub:
```

```
# install.packages("devtools")
```

```
devtools::install_github("tidyverse/tidyr")
```

Get the cheatsheet at [here](#)

1. 长宽数据转变: 宽数据 向 长数据 转变

get data ready

```
library(tidyverse); ## 先装入包;
grades2 <- read_tsv(file = "data/talk06/grades2.txt");

grades2;
```

A tibble: 3 x 6

	name	Microbiology	English	Chinese	Bioinformatics	Chemistry
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Zhi Liu	100	50	69	NA	NA
2	Weihua Chen	89	99	NA	99	NA
3	Kang Ning	NA	NA	20	100	76

宽数据的特点

优点:

- 自然, 易理解;

缺点:

- 不易处理;
- 稀疏时问题较大;

宽数据 向 长数据 转变

```
library(kableExtra);
grades3 <- grades2 %>% pivot_longer( - name, names_to = "course", values_to = "grade" );
kbl( grades3 );
```

name	course	grade
Zhi Liu	Microbiology	100
Zhi Liu	English	50
Zhi Liu	Chinese	69
Zhi Liu	Bioinformatics	NA
Zhi Liu	Chemistry	NA
Weihua Chen	Microbiology	89
Weihua Chen	English	99
Weihua Chen	Chinese	NA
Weihua Chen	Bioinformatics	99
Weihua Chen	Chemistry	NA
Kang Ning	Microbiology	NA
Kang Ning	English	NA
Kang Ning	Chinese	20
Kang Ning	Bioinformatics	100
Kang Ning	Chemistry	76

pivot_longer explained!

```
grades3 <- grades2 %>% pivot_longer( - name, names_to = "course", values_to = "grade" );
```

-name: 此列保留

列名变为第一列, 取名为 course

name	Bioinformatics	Chemistry	Chinese	English	Microbiology
Kang Ning	100	76	20	NA	NA
Wei-hua Chen	99	NA	NA	99	89
Zhi Liu	NA	NA	69	50	100

值变为第二列, 取名为 grade

Figure 2: pivot_longer explained!

有 NA 值怎么办？

```
grades3_1 <- grades3[ !is.na(grades3$grade), ];  
grades3_2 <- grades3[ complete.cases( grades3 ) , ];
```

-- 更好的方法 ~~

```
grades3_long <- grades2 %>%  
  pivot_longer( - name,  
               names_to = "course",  
               values_to = "grade",  
               values_drop_na = TRUE);
```

values_drop_na 即可消除;

有 NA 值怎么办? cont.

```
kbl( grades3_long );
```

name	course	grade
Zhi Liu	Microbiology	100
Zhi Liu	English	50
Zhi Liu	Chinese	69
Weihua Chen	Microbiology	89
Weihua Chen	English	99
Weihua Chen	Bioinformatics	99
Kang Ning	Chinese	20
Kang Ning	Bioinformatics	100
Kang Ning	Chemistry	76

长变宽

```
grades3_wide <- grades3_long %>%
  pivot_wider( names_from = "course", values_from = "grade" );

grades3_wide;
```

A tibble: 3 x 6

	name	Microbiology	English	Chinese	Bioinformatics	Chemistry
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Zhi Liu	100	50	69	NA	NA
2	Weihua Chen	89	99	NA	99	NA
3	Kang Ning	NA	NA	20	100	76

pivot_wider 怎么用?

1. 选取哪列做“列名”



name	course	grade
Zhi Liu	Microbiology	100
Zhi Liu	English	50
Zhi Liu	Chinese	69
Weihua Chen	Microbiology	89
Weihua Chen	English	99
Weihua Chen	Bioinformatics	99
Kang Ning	Bioinformatics	100
Kang Ning	Chinese	20
Kang Ning	Chemistry	76

2. 选取哪列
做“数据”



Figure 3: pivot_wider function explained

宽长数据转换练习

用 `pivot_wider` 和 `pivot_longer` 对下面的数据 `mini_iris` 进行宽长转换:

```
mini_iris <- iris[ c(1, 51, 101),  ];
kbl( mini_iris);
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
51	7.0	3.2	4.7	1.4	versicolor
101	6.3	3.3	6.0	2.5	virginica

`iris` 是 鸢尾属 一些物种花瓣的量表

宽变长, cont.

```
## -- 注意: 第一、二个参数可以自行命名, 分别对应原始数据中的 column names 及 values ...
mini_iris.longer <- mini_iris %>%
  pivot_longer( - Species, names_to = "type", values_to = "dat" );
kbl( mini_iris.longer );
```

Species	type	dat
setosa	Sepal.Length	5.1
setosa	Sepal.Width	3.5
setosa	Petal.Length	1.4
setosa	Petal.Width	0.2
versicolor	Sepal.Length	7.0
versicolor	Sepal.Width	3.2
versicolor	Petal.Length	4.7
versicolor	Petal.Width	1.4
virginica	Sepal.Length	6.3
virginica	Sepal.Width	3.3
virginica	Petal.Length	6.0
virginica	Petal.Width	2.5

长变宽

```
## -- 注意: 第一、二个参数可以自行命名, 分别对应原始数据中的 column names 及 values ...
mini_iris.wider <- mini_iris.longer %>%
  pivot_wider( names_from = "type", values_from = "dat" );
kbl( mini_iris.wider );
```

Species	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
setosa	5.1	3.5	1.4	0.2
versicolor	7.0	3.2	4.7	1.4
virginica	6.3	3.3	6.0	2.5

比较复杂的例子

```
grades2 <- read_delim( file = "data/talk05/grades2.txt", delim = "\t",
                        quote = "", col_names = T);
kbl( grades2 );
```

name	class	course	grade
CHEN	1	bioinformatics	90
CHEN	1	chemistry	92
CHEN	2	chinese	35
CHEN	3	german	62
LI	1	bioinformatics	44
LI	2	chinese	68
LI	3	microbiology	95
LI	3	japanese	90
WANG	1	bioinformatics	35
WANG	1	chemistry	76
WANG	1	mathmatics	82
WANG	3	german	100
WANG	3	spanish	78

这是哪种数据类型？长还是宽??

怎么变成宽数据?

```
grades2_wide <- grades2 %>%
  pivot_wider( names_from = course, values_from = grade );

kbl( grades2_wide );
```

name	class	bioinformatics	chemistry	chinese	german	microbiology	japanese	mathmatics	sp
CHEN	1	90	92	NA	NA	NA	NA	NA	
CHEN	2	NA	NA	35	NA	NA	NA	NA	
CHEN	3	NA	NA	NA	62	NA	NA	NA	
LI	1	44	NA	NA	NA	NA	NA	NA	
LI	2	NA	NA	68	NA	NA	NA	NA	
LI	3	NA	NA	NA	NA	95	90	NA	
WANG	1	35	76	NA	NA	NA	NA	82	
WANG	3	NA	NA	NA	100	NA	NA	NA	



再变成长数据

又怎么把它变回来？

```
a <-
  grades2_wide %>%
  pivot_longer( ! c( name, class ),
                names_to = "course",
                values_to = "grade",
                values_drop_na = T
              );

kbl( a );
```

name	class	course	grade
CHEN	1	bioinformatics	90
CHEN	1	chemistry	92
CHEN	2	chinese	35
CHEN	3	german	62
LI	1	bioinformatics	44
LI	2	chinese	68
LI	3	microbiology	95
LI	3	japanese	90
WANG	1	bioinformatics	95

另一种变法，注意两者的区别！！

```
b <- grades2_wide %>%
  pivot_longer( bioinformatics:spanish, ## 选择成绩所在的列！
               names_to = "course", values_to = "grade",
               values_drop_na = T
             );

kbl( b );
```

name	class	course	grade
CHEN	1	bioinformatics	90
CHEN	1	chemistry	92
CHEN	2	chinese	35
CHEN	3	german	62
LI	1	bioinformatics	44
LI	2	chinese	68
LI	3	microbiology	95
LI	3	japanese	90
WANG	1	bioinformatics	35
WANG	1	chemistry	76
WANG	1	mathmatics	82
WANG	3	german	100

2. tidyr::separate 将一列拆成多列

```
table3 <- read_tsv(file = "data/talk06/table3.txt");

table3 %>%
  separate(rate, into = c("cases", "population"), sep = "/");
```

A tibble: 6 x 4

	country	year	cases	population
	<chr>	<dbl>	<chr>	<chr>
1	Afghanistan	1999	745	19987071
2	Afghanistan	2000	2666	20595360
3	Brazil	1999	37737	172006362
4	Brazil	2000	80488	174504898
5	China	1999	212258	1272915272
6	China	2000	213766	1280428583

tidyr::separate 同时进行格式转换

如何把分拆后的列正确识别为数字??

```
table3 %>%
  separate(rate, into = c("cases", "population"), convert = TRUE)
```

```
# A tibble: 6 x 4
  country      year cases population
  <chr>      <dbl> <int>      <int>
1 Afghanistan 1999     745    19987071
2 Afghanistan 2000    2666    20595360
3 Brazil       1999   37737   172006362
4 Brazil       2000   80488   174504898
5 China        1999  212258  1272915272
6 China        2000  213766  1280428583
```

tidyr::separate 按字符串长度分割

把年拆分为世纪和年

```
table5 <- table3 %>%
  separate(year, into = c("century", "year"), sep = 2)

table5;
```

```
# A tibble: 6 x 4
  country    century year    rate
  <chr>      <chr>   <chr> <chr>
1 Afghanistan 19      99    745/19987071
2 Afghanistan 20      00    2666/20595360
3 Brazil      19      99    37737/172006362
4 Brazil      20      00    80488/174504898
5 China       19      99    212258/1272915272
6 China       20      00    213766/1280428583
```

3. tidyr::unite 将多列合成一行

将上页分拆的结果进行合并

```
table5 %>%
  unite(new, century, year, sep = ""); ## sep 参数默认是 _
```

A tibble: 6 x 3

	country	new	rate
	<chr>	<chr>	<chr>
1	Afghanistan	1999	745/19987071
2	Afghanistan	2000	2666/20595360
3	Brazil	1999	37737/172006362
4	Brazil	2000	80488/174504898
5	China	1999	212258/1272915272
6	China	2000	213766/1280428583

seperate 与 unit 小结

- 分割和合并后, 原列会消失! 可使用 `remove = FALSE` 保留原列
- 更多示例见: <https://r4ds.had.co.nz/tidy-data.html>

4. 其它函数

- fill (作业会用到)
- complete (作业会用到)
- ...

更多示例见: <https://r4ds.had.co.nz/tidy-data.html>

Section 3

section 3 : 小结与作业

小结

今次提要

- tidyr (超级强大的数据处理) part 2

下次预告

- dplyr, tidyr 和 forcats 的更多功能与生信操作实例

important

- all codes are available at Github:
<https://github.com/evolgeniusteam/R-for-bioinformatics>

练习 & 作业

- 暂无;